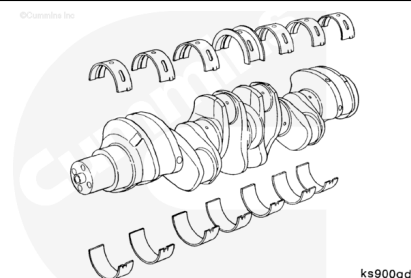


Trainings ▾

General Information

The crankshaft uses forged counterweights.

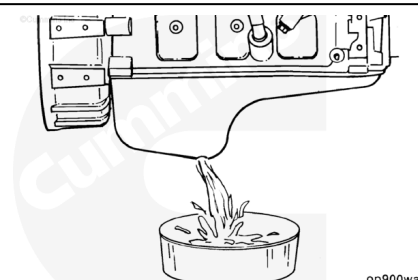
Oversize main bearings and thrust bearings are available for service. Cummins Engine Company, Inc., recommends regrounding **all** of the main or the connecting rod journals when one requires regrounding. Refer to the C Series Shop Manual, Bulletin No. 3666008.



Preparatory Steps

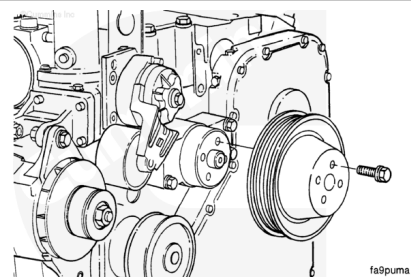
⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.

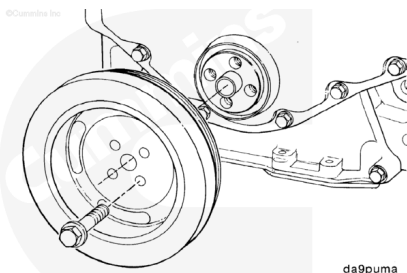


Drain the lubricating oil; refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/100/100-007-037.html>).

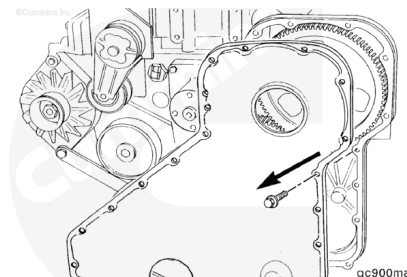
Remove the fan pulley; refer to Procedure 008-039 (</qs3/pubsys2/xml/en/procedures/100/100-008-039.html>).



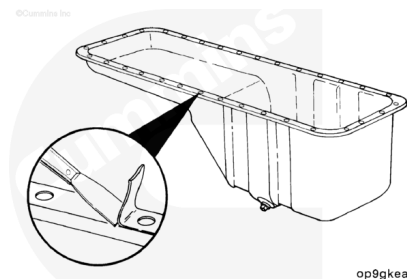
Remove the vibration damper; refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/100/100-001-052.html>).



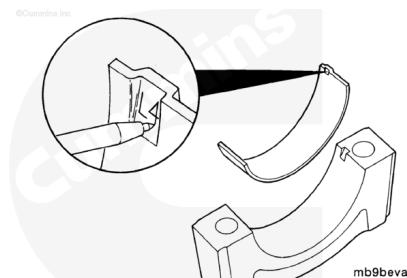
Remove the front gear cover; refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/100/100-001-031.html>).



Remove the lubricating oil pan; refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/100/100-007-025.html>).



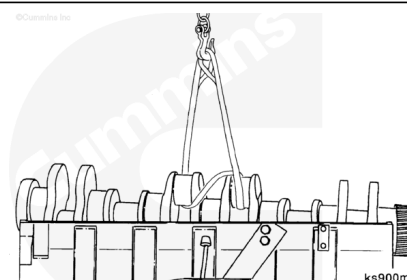
Remove the main bearings; refer to Procedure 001-006 (</qs3/pubsys2/xml/en/procedures/100/100-001-006.html>).



Remove

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this component.



Note : Lift the crankshaft straight up to avoid damage to the crankshaft and cylinder block.

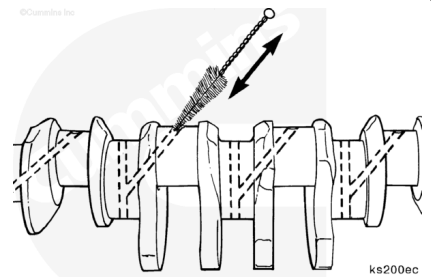
Install nylon lift sling, Part No. 3375957, around the No. 3 and No. 4 rod bearing journals.

Attach the sling to a hoist; remove the crankshaft.

Clean

Use fine crocus cloth to polish the machined surfaces.

Use a bristle brush to clean the oil drillings.

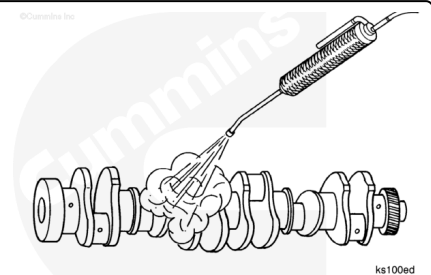


⚠ WARNING ⚠

When using a steam cleaner, wear protective clothing and safety glasses or a face shield. Hot steam will cause serious personal injury.

Steam-clean the crankshaft and dry with compressed air.

Make sure to blow out the threaded holes on each end of the crankshaft and the oil drillings.



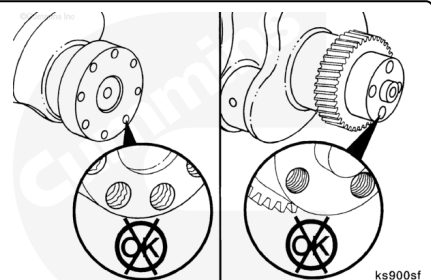
Inspect for Reuse

Inspect the threaded capscrew holes for damage.

Use one of the following methods to repair any threaded holes:

- Chase the threads
- Use the threaded insert kit, Part Number 3822709.

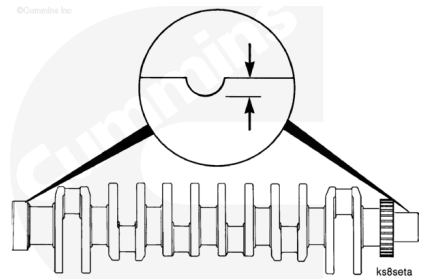
Note : If more than two threaded holes are damaged in one end, the crankshaft must be replaced.



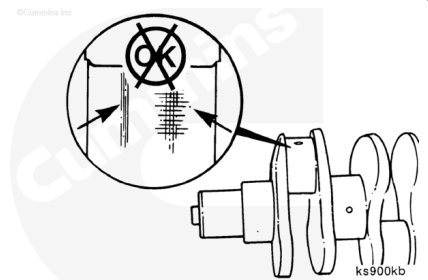
Measure the front and rear oil seal contact areas for a wear groove.

mm		in
0.25	MAX	0.010

Front and rear wear sleeves are available for a crankshaft that is worn beyond the limit.

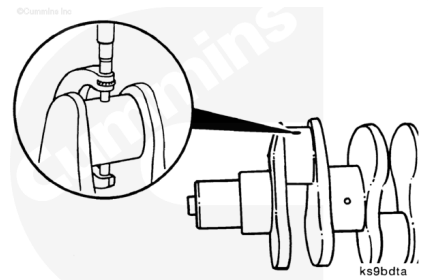


Inspect the crankshaft connecting rod and main journals for deep scoring, overheating, etc.



Use a micrometer to measure the connecting rod journal.

	mm		in
Standard	75.96	MIN	2.991
Machined 0.25 mm 0.010 in	75.71	MIN	2.981
Machined 0.50 mm 0.020 in	75.46	MIN	2.971
Machined 0.75 mm 0.030 in	75.21	MIN	2.961
Machined 1.00 mm 0.040 in	74.96	MIN	2.951

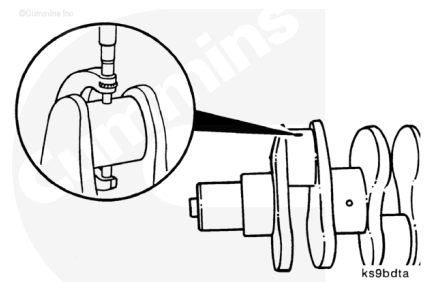


Bearing Clearance = Connecting rod inside diameter with bearing installed minus crankshaft journal diameter.

Maximum Bearing Clearance: 0.117 mm [0.0046 in].

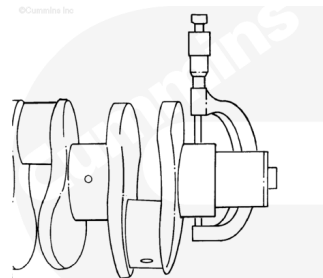
	mm		in
Out-of-Roundness		0.050	MAX
			0.0020

Taper		0.013	MAX	0.0005
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ks9bdtA

	mm		in
Standard	97.96	MIN	3.857
Machined 0.25 mm 0.010 in	97.71	MIN	3.847
Machined 0.50 mm 0.020 in	97.46	MIN	3.837

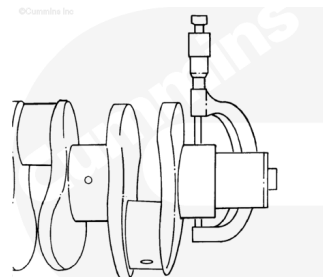


ks900nc

Bearing Clearance = Main bore diameter with bearing installed minus (-) crankshaft main journal diameter.

Maximum Bearing Clearance: 0.13 mm [0.005 in].

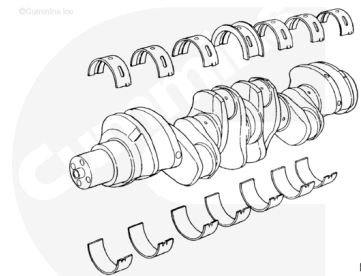
	mm		in	
Out-of-Roundness		0.05	MAX	0.002
Taper		0.01	MAX	0.001



ks900nc

The following oversize service main and thrust bearings are available for crankshafts that have been machined undersize:

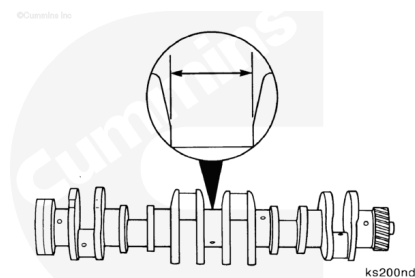
Thrust Size				
	Oversize	0.000	0.25 mm	0.50 mm
Journal	0.000	X	X	X
Diameter	0.25 mm	X	X	
	0.50 mm	X		X
	0.75 mm	X		
	1.00 mm	X		



ks900gd

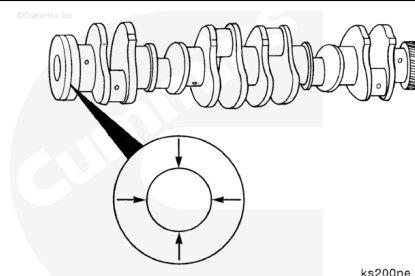
Measure the thrust face width.

mm		in
42.98	MIN	1.692
43.08	MAX	1.696



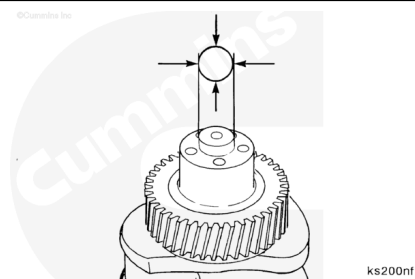
Measure the rear oil seal flange outside diameter.

mm		in
129.98	MIN	5.117
130.03	MAX	5.119



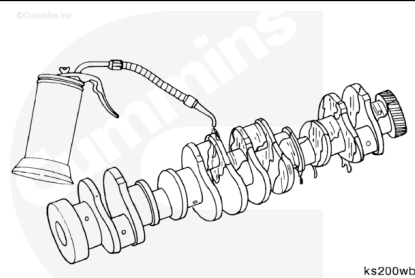
Measure the damper pilot outside diameter.

mm		in
23.92	MIN	0.942
24.00	MAX	0.945

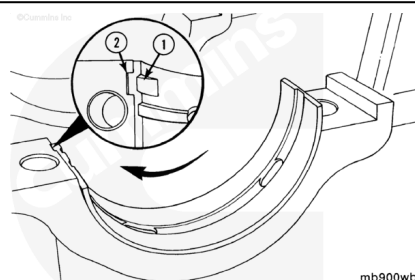


Use a light preservative oil to lubricate the crankshaft to prevent rust.

Note : If the crankshaft is not going to be used immediately, protect the part with a plastic cover to prevent dirt from sticking to the oil.

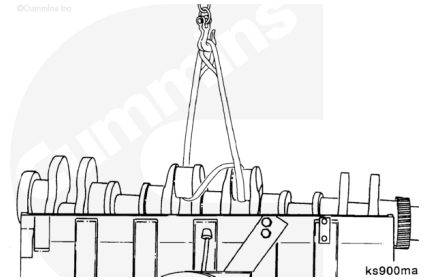


Install the upper main bearings; refer to Procedure 001-006 (/qs3/pubsys2/xml/en/procedures/100/100-001-006.html).



⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this component.

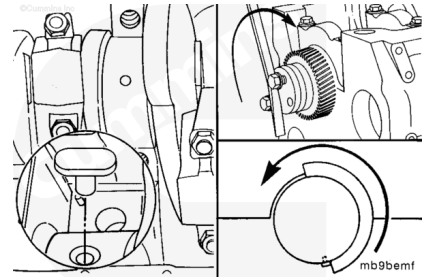


Use a hoist and nylon lift sling, Part No. 3375957.

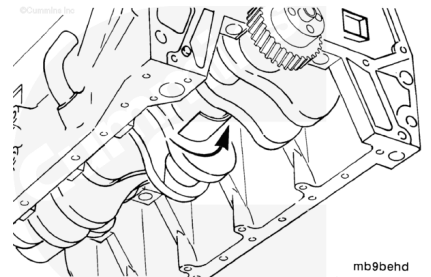
Install the sling around the No. 3 and No. 4 connecting rod bearing journals.

Install the crankshaft.

Install the upper thrust bearing; refer to Procedure 001-006 ([/qs3/pubsys2/xml/en/procedures/100/100-001-006.html](http://qs3/pubsys2/xml/en/procedures/100/100-001-006.html)).

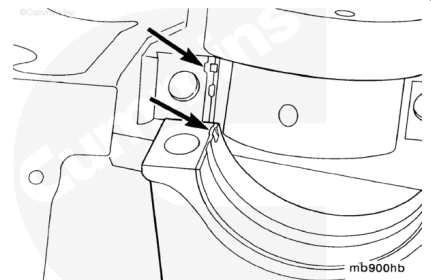


Install the lower main bearing; refer to Procedure 001-006 ([/qs3/pubsys2/xml/en/procedures/100/100-001-006.html](http://qs3/pubsys2/xml/en/procedures/100/100-001-006.html)).

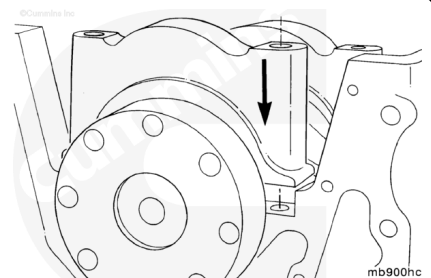


Note : The main bearing caps are numbered for location. No. 1 starts with the front of the cylinder block and the numbers must face the camshaft side of the engine.

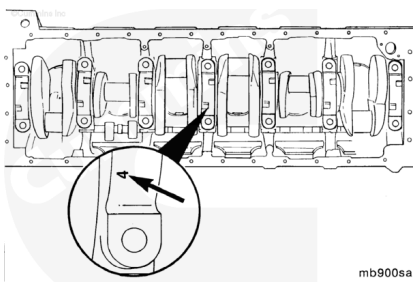
When installed correctly, the main bearing tangs will both be on the same side.



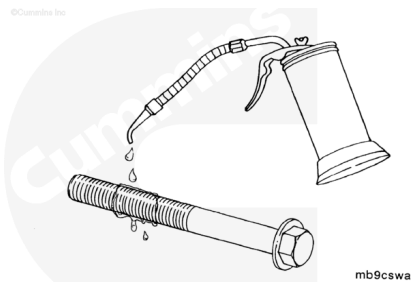
Position the main bearings and main bearing caps onto the crankshaft.



The thrust bearing journal does **not** have tangs or slots. Care **must** be taken to make sure the number stamped on the main bearing cap is positioned to the camshaft side of the engine.



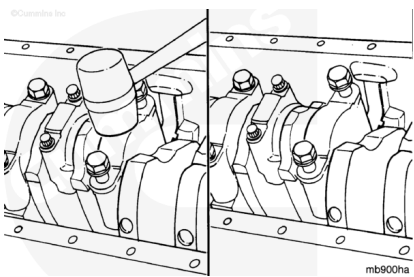
Lubricate the main bearing capscrew threads and underside of the capscrew head with clean engine oil.



Tap the main bearing cap gently into position.

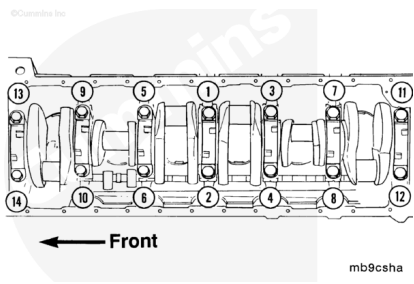
Note : Make sure the main bearing is still aligned with the cap.

When seated, the main bearing capscrews can be threaded in by hand.



Tighten the main bearing capscrews evenly following the illustrated sequence.

Torque Value:	Step 1	50 n.m [37 ft-lb]
	Step 2	95 n.m [70 ft-lb]
	Step 3	60°



Note : The crankshaft must rotate freely after the main bearings have been installed.

Inspect the main bearing installations and the size of the main bearings if the crankshaft will **not** rotate freely.

